

TED (15)4072

MODEL QUESTION

**FOURTH SEMESTER DIPLOMA EXAMINATION IN CHEMICAL
ENGINEERING**

HEAT TRANSFER AND EVAPORATION

Time: 3hrs

Maximum marks: 100

PART –A

Maximum marks: 10

I Answer the following questions in one or two sentences. Each question carries 2 marks

- 1 Define conduction.
- 2 Define emissivity.
- 3 State SteafanBotzmann Law.
- 4 Define baffle pitch.
- 5 State Duhring rule. (5X2=10)

PART –B

(Maximum marks:30)

II Answer any five of the following questions. Each question carries 6 marks.

- 1 Find the rate of heat loss through a wall of red brick 6
($k=0.7\text{W/mK}$) 5m in length 4m in height and 250mm in

thickness. If wall surface is maintained at 373K and 303K respectively.

- | | | |
|---|--|---|
| 2 | Derive an equation for conduction through cylindrical wall. | 6 |
| 3 | Differentiate between Natural convection and Forced convection with example. | 6 |
| 4 | Draw the neat sketch of Fixed tube heat exchanger and label its parts. | 6 |
| 5 | A hot fluid enters a double pipe heat exchanger at a temperature of 423K and is to be cooled to 367K. A cold fluid entering at 311K and heated to 339K. Compute LMTD in parallel flow. | 6 |
| 6 | Draw the neat sketch of standard vertical tube evaporator and label its parts. | 6 |
| 7 | Summarize Backward feed arrangement in multiple effect evaporation system. | 6 |
- (5x6=30)

PART – C

Maximum marks:60

(Answer one full question from each unit. Each full question carries 15 marks)

Module -I

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|-----|-----|--|---|
| III | (a) | A furnace is constructed with 25cm thick of fire clay brick, 10cm of diatomaceous earth brick and 5cm of the common brick. The inside temperature is 1743K and outside temperature is 373K. Find the heat loss per unit area and the temperature of the junctions of the layers. Thermal conductivity of fireclay brick=1.09w/(mk)
Thermal conductivity of diatomaceous earth brick =0.26w/(mk) | 9 |
|-----|-----|--|---|

Thermal conductivity of common brick = 0.65 W/(mK)

- (b) Derive an equation for conduction through a spherical wall. 6

OR

- IV (a) Explain temperature gradient in forced convection. 7

- (b) A steam pipe line 150mm/160mm in diameter is covered with a layer of insulating material of thickness 100mm. The temperature inside of the pipeline is 392.8K and that of the outside surface of insulation is 313K. Determine the rate of heat transfer per metre length of pipe line. 8

Thermal conductivity of insulating material = 0.08 W/(mK)

Module II

- V (a) Formulate the dimensionless equations in heat transfer. 5
- (b) Find the inside heat transfer coefficient for fluid flowing at a rate of $300 \times 10^{-6} \text{ m}^3/\text{s}$ through 30mm inside diameter tube heat exchanger. 10

DATA

Density of flowing fluid = 1100 Kg/m^3

Viscosity of flowing fluid = $0.7 (\text{Ns})/\text{m}^2$

Specific heat of fluid = 1.22 KJ/KgK

Thermal conductivity of fluid = 0.385 W/mK

Viscosity at wall temperature = $1 (\text{Ns})/\text{m}^2$

Length of heat exchanger = 6M

OR

- VI (a) Justify Kirchhoff law. 6
- (b) Find the rate of heat transfer by radiation from an 6

unlagged steam pipe of 60mm OD at 400K to air at 300K. Assume emissivity = 0.7.

- (c) Distinguish between Opaque body and gray body 3

Module III

- VII (a) Explain the constructional details and working of double pipe heat exchanger. 6

- (b) Water enters a two fluid heat exchanger at 328K and leaves at 358K. Hot gas enter at 578K and leaves at 433K. If total heat transfers Area 600m^2 and the overall heat transfer coefficient are $800\text{ w /m}^2\text{K}$. Determine the total heat transferred for parallel flow and counter current flow. 9

OR

- VIII (a) A heat exchanger is required to cool 25Kg/s of water from 380K to 340K by means of 35Kg/s of water entering at 300K. If the overall heat transfer coefficient is $4000\text{w/m}^2\text{K}$. Compute surface area required in parallel flow. Cp of water = 4.187KJ/KgK 9
- (b) Explain the constructional details and working of Fin type heat exchangers. 6

Module IV

- IX (a) Illustrate the working of Basket type evaporator. 6
- (b) A single effect evaporator is to concentrate 30000Kg/hr of solution having concentration of 10% salt to a concentration of 50% salt by weight. Steam is 9

feed to evaporator at a pressure corresponding to saturation temperature of 399K. The evaporator is working at reduced pressure such that boiling point is 323K. Overall heat transfer coefficient is $2.9 \text{ KW/m}^2\text{K}$. Compute (1) Capacity (2) Economy (3) Steam consumption (4) Heat transfer Area.

Feed temperature = 293K.

Specific heat of Feed = 3.98 KJ/KgK .

Latent heat of steam at 399K = 2202 KJ/Kg .

Latent heat of vapourisation of water at 373K = 2383 KJ/Kg

OR

X (a) Summarize the working of Steam jet compression 5
evaporator.

(b) A single effect evaporator is fed with 2000Kg/hr of 10
solution containing 10% solute by weight. Feed
temperature is 313K and is to be concentrated to a
solution 40% by weight. The evaporator is at
atmospheric pressure. Saturated steam is supplied at
303.97KPa saturation temperature of 407K as a
heating medium. Heat transfer coefficient is
 $1750 \text{ W/m}^2\text{K}$. Compute (1) capacity (2) economy (3)
Heat load and (4) heat transfer area.

DATA

Hf of feed at 313K = 170 KJ/Kg

Hs = 2775 KJ/Kg

Hv = 2676 KJ/Kg

Boiling point solution = 373K

H = 419 KJ/Kg

Hc = 563 KJ/Kg

